TrES-5b

R-1793

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_210814` · created 2026-08-03T01:01:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459440.8711 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.058 +/- 0.052
Transit depth (Rp/Rs)^2	0.34 +/- 0.61 [%]
Orbital Inclination (inc)	80.31 +/- 2.91
Airmass coefficient 1 (a1)	1.79 +/- 0.77
Airmass coefficient 2 (a2)	-0.47 +/- 0.32
Scatter in the residuals of the lightcurve fit is	41.99 %
Best Comparison Star	#1 - [302, 370]
Optimal Aperture	3.07
Optimal Annulus	7.42
Transit Duration (day)	0.025 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.058 ± 0.052 vs 0.142 (deviation 1.62σ)
Duration fit vs published	0.025 d vs 0.0758 d (deviation 1.82σ)
Mid-time O–C	-1.8 min
Depth SNR	1.1
Residual scatter	41.99 %

Light curve

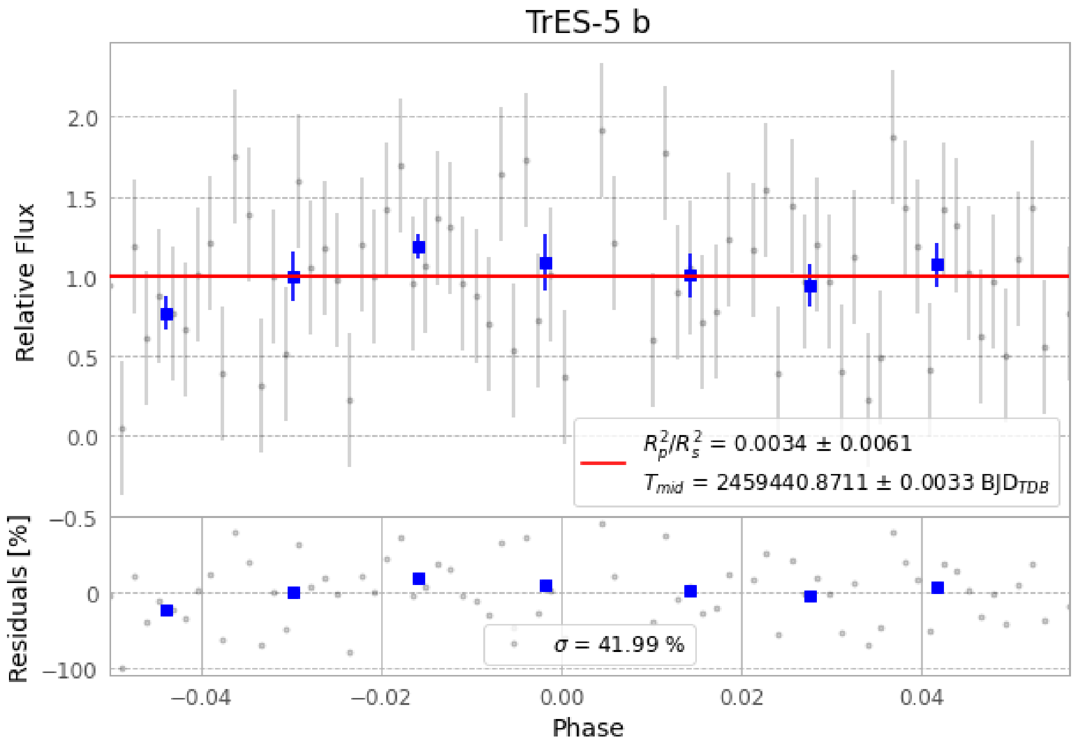

Figure 1 — FinalLightCurve_TrES-5 b_2021-08-14.png (SHA-256 493789ba4cea1182…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 434], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6e1f75602aabaa16…).
`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

75 FITS frames (49 MB), `TRES-5210814070312.FITS` → `TRES-5210814105112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-210814.log (15277c125eae…), FinalLightCurve_TrES-5 b_2021-08-14.png (493789ba4cea…), FinalLightCurve_TrES-5 b_2021-08-14.csv (902e007c0f56…)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)